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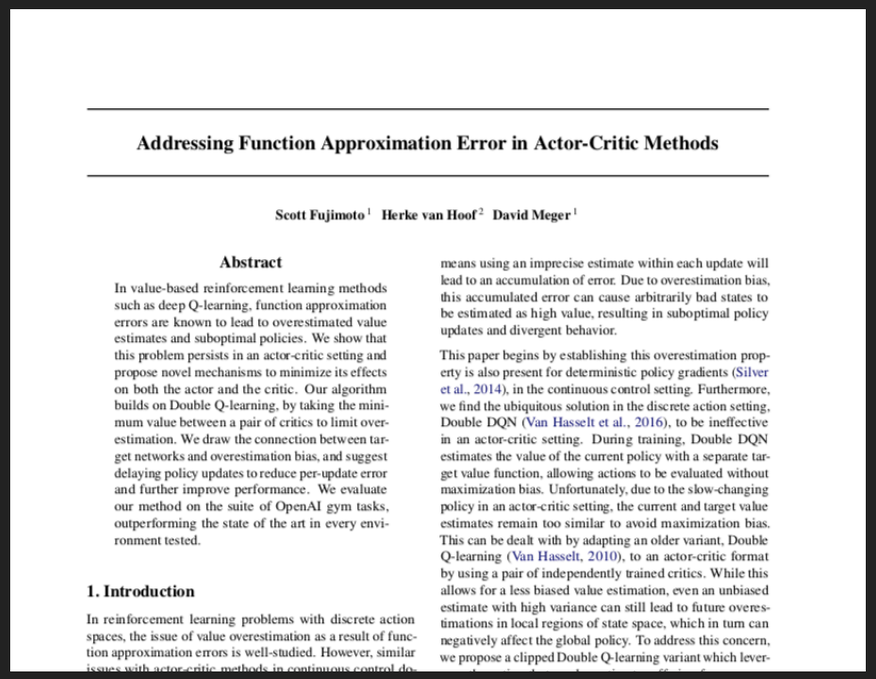
TD3

INTRODUCCIÓN

- TD3 es un algoritmo RL diseñado para resolver tareas continuas que mejora la estabilidad y rendimiento del DDPG.

INTRODUCCIÓN

- TD3 significa Twin Delayed Deep Deterministic Policy Gradient.
- Fue publicado en 2018 por Fujimoto y colegas bajo el título "Addressing Function Approximation Error in Actor-Critic Methods".



Addressing Function Approximation Error in Actor-Critic Methods

Scott Fujimoto [1] Herke van Hoof [2] David Meger [1]

Abstract

In value-based reinforcement learning methods such as deep Q-learning, function approximation errors are known to lead to overestimated value estimates and suboptimal policies. We show that this problem persists in an actor-critic setting and propose novel mechanisms to minimize its effects on both the actor and the critic. Our algorithm builds on Double Q-learning, by taking the minimum value between a pair of critics to limit overestimation. We draw the connection between target networks and overestimation bias, and suggest delaying policy updates to reduce per-update error and further improve performance. We evaluate our method on the suite of OpenAI gym tasks, outperforming the state of the art in every environment tested.

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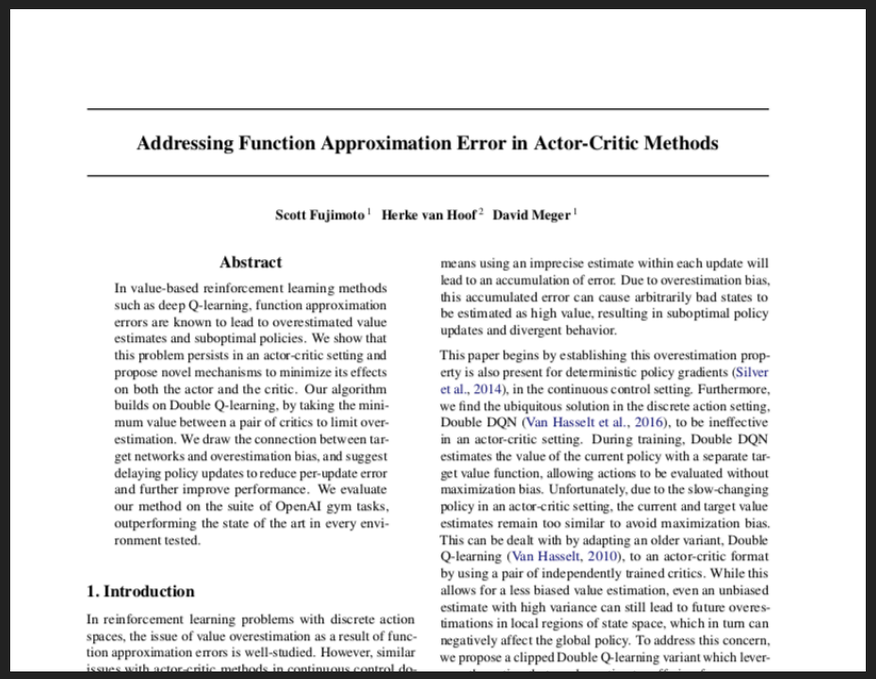
This paper begins by establishing this overestimation property is also present for deterministic policy gradients (Silver et al., 2014), in the continuous control setting. Furthermore, we find the ubiquitous solution in the discrete action setting, Double DQN (Van Hasselt et al., 2016), to be ineffective in an actor-critic setting. During training, Double DQN estimates the value of the current policy with a separate target value function, allowing actions to be evaluated without maximization bias. Unfortunately, due to the slow-changing policy in an actor-critic setting, the current and target value estimates remain too similar to avoid maximization bias. This can be dealt with by adapting an older variant, Double Q-learning (Van Hasselt, 2010), to an actor-critic format by using a pair of independently trained critics. While this allows for a less biased value estimation, even an unbiased estimate with high variance can still lead to future overestimations in local regions of state space, which in turn can negatively affect the global policy. To address this concern, we propose a clipped Double Q-learning variant which lever-

¿POR QUÉ SE CREÓ TD3?

- El DDPG presenta un problema, sobreestima los valores de las acciones, lo que llevaba a malas decisiones y entrenamientos inestables.



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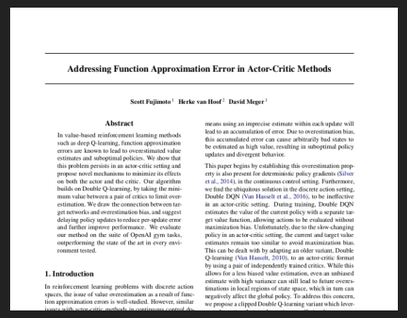
¿COMO SOLUCIONA TD3 ESE PROBLEMA?

- 1. Twin (Gemelos) → Dos críticos (Q-networks)
  - ✓ En lugar de tener una red que estima cuán buena es una acción (llamada "crítico" o "Q-function"), TD3 usa dos redes críticas.
  - ✓ Al tomar una decisión, elige el valor Q más bajo de las dos.
  - ✓ Esto evita la sobreestimación, si una red dice "esta acción vale 100" y la otra dice "vale 80", se usa entonces 80 (más conservador).
  - ✓ Esto se llama "clipped double Q-learning".



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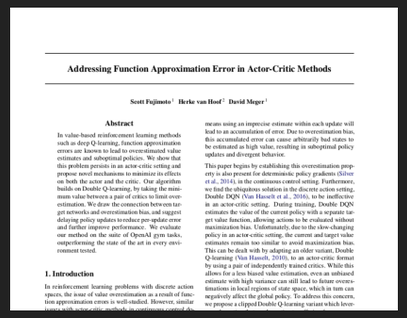
- 2. Delayed (Retrasado) → Actualiza la política menos seguido
  - ✓ La política decide qué acción debe tomar el actor.
  - ✓ En DDPG se actualiza la política en cada paso lo que puede llevar a errores acumulados.
  - ✓ En TD3 se actualiza la política (actor) menos frecuentemente que los críticos.
  - ✓ Por ejemplo: se actualiza los críticos 2 veces, y luego se actualiza la política 1 vez.
  - ✓ Esto da más tiempo a los críticos para "aprender bien" antes de guiar al actor.



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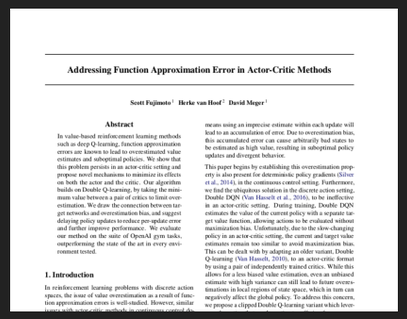
- 3. Target Policy Smoothing → Suavizado de la política objetivo
  - ✓ Cuando el agente evalúa acciones futuras en DDPG se usa la mejor acción directamente.
  - ✓ Eso puede llevar a "picos" o acciones muy específicas que no generalizan bien.
  - ✓ TD3 añade un poco de ruido a la acción objetivo y luego la "suaviza" (promedia con acciones cercanas).
  - ✓ Así, el agente aprende a ser más robusto y no depende de acciones ultra-específicas.



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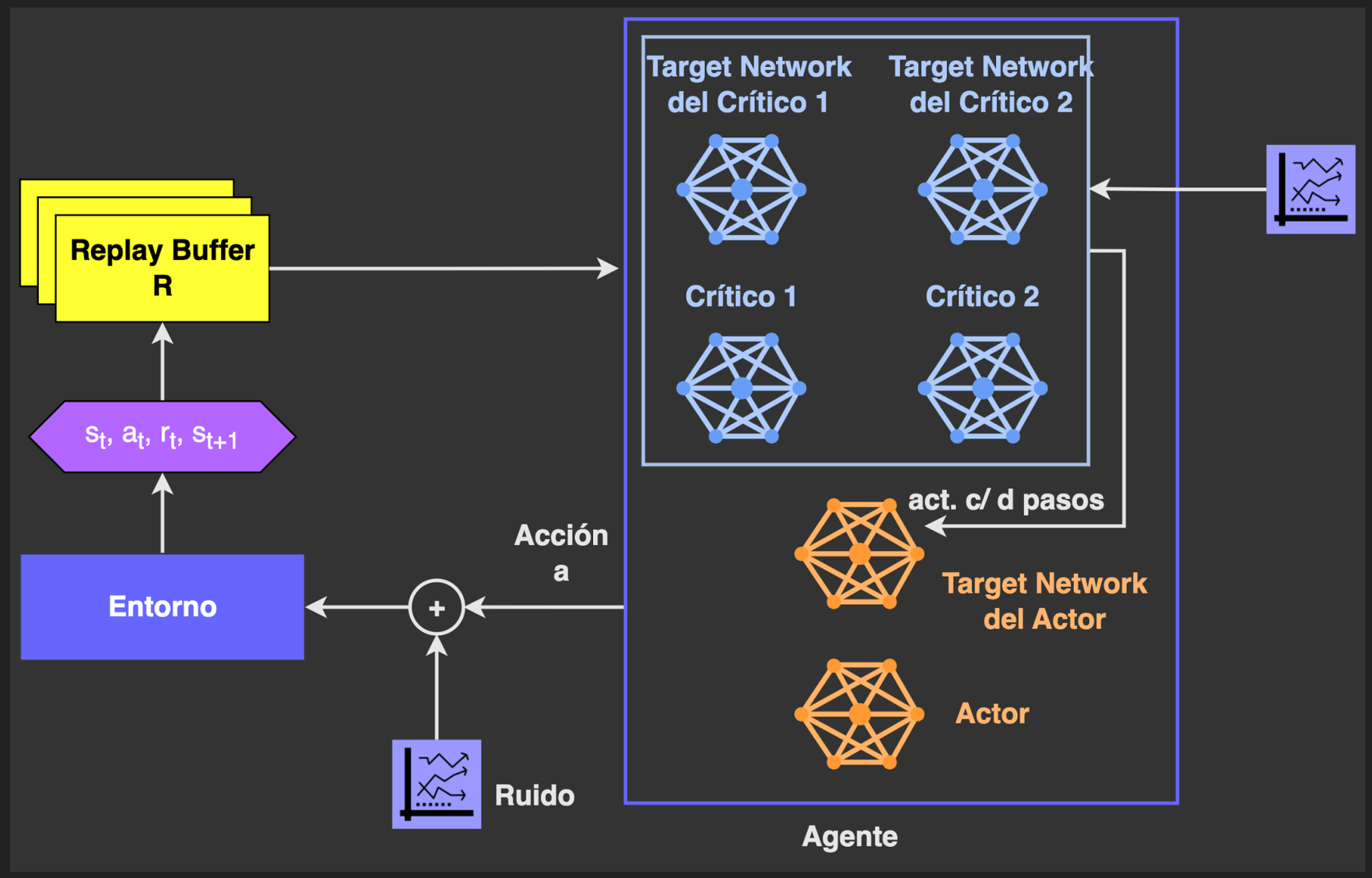

REFERENCIAS BIBLIOGRÁFICAS Y WEB (I)

- Fujimoto, S., Hoof, H., & Meger, D. (2018, July). Addressing function approximation error in actor-critic methods. In International conference on machine learning (pp. 1587-1596). PMLR.